Large area crack-free GaN on top of Si-on-insulator substrates

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ABSTRACT

GaN was grown by metal organic chemical vapor deposition (MOCVD) on plain Si on insulator (SOI) substrates, and on SOI which was pre-patterned for lateral confined epitaxy (LCE), with square mesas, separated by dip trenches. It was found that GaN films grown on plain SOI exhibit reduced crack density. Moreover, using LCE on SOI substrates dramatically increases the crack-free area of the layer. Areas of 100 μm × 100 μm were grown completely crack free, and stress was released by strong bending of the GaN/Si/SiO_2 composite islands.

Keywords: GaN on Si, Lateral Confined Epitaxy, Si-on-Insulator.

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GaN layers grown on plain Si(111) substrates by metalorganic chemical vapor deposition (MOCVD) exhibit dense cracking along $\{1\,\bar{1}\,00\}$ planes, due to large tensile stress.[1] The stress originates from the large difference in the thermal expansion coefficient between GaN and Si (56%). Cracking occurs during the cooling after the growth, since the substrate/layer structure bends upwards, inducing large tensile stress in the layer. Cracks cross the area randomly, and induce non-uniformity in material properties.

One way to alleviate the thermal crack problem is by introducing several intermediate layers[2] or a graded buffer layer[3] for strain compensation. However, this method imposes the requirement that the GaN epilayer is grown under highly compressive strain, thus compromising material quality.[3]

To circumvent this limitation an alternative approach is *substrate engineering* (i.e. reducing the substrate size in lateral and/or vertical dimension). By this method the built-in strain in the layer is diluted, and partially transferred to the substrate.

The reduction in density of thermal cracks by decreasing the *lateral* size of the substrate growth units (mesas) was studied previously by Lateral Confined Epitaxy (LCE).[4, 5] It was shown that for mesas, of less than ~14 μm in width, a GaN layer ~0.7 μm thick can grow with no cracks. This maximum crack free area was also verified with thicker GaN films, concluding that 13-14 μm can be considered as the characteristic length scale for the crack free growth of GaN on Si(111).[6] However, this crack-free lateral size is one order of magnitude smaller than an average optoelectronic device.

In this work we, we study the effect of substrate thickness reduction (*vertical* dimension) on the crack-free length scale of GaN on Si(111). By using LCE of GaN on Silicon-on-Insulator (SOI), it is shown that when the substrate thickness is comparable to that of the epilayer, the crack-free length scale is increased to above 100 μm. The strain release mechanism results in strong bowing of the substrate-epilayer composite.

The growth of GaN on compliant substrates has been discussed before, and in particular the growth on SOI substrates.[7, 8, 9] This substrate contains a thick bottom substrate (handle) and a thin top Si layer (device layer) attached by a buried SiO_2 (BOX) layer (the insulator). The "device layer" is used as a compliant substrate for the epitaxial growth of GaN, since it has two main special properties: (1) the small

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substrate thickness, which reduces the stresses induced in the GaN by bending, and (2) it is located on a flexible oxide, which may absorb a part of the strain.

In spite of the expected compliance of SOI substrates, it is important to determine whether simply using SOI as a substrate instead of plain Si can completely eliminate the formation of thermal cracks, or whether additional substrate treatment, such as the use of LCE, is required. To be more general: is a vertical confinement of the substrate thickness enough, or there is a need to confine the growth laterally as well?

Commercial SOI wafers of BCO-Technologies with device Si(111) layer, BOX and Si(100) handle of thicknesses ~2.3 μm, ~2 μm, and ~450 μm, respectively,[10] were used as substrates. GaN was grown by low pressure (100 Torr) MOCVD on top of the device layer, using a ~50 nm AlN as nucleation layer. For this experiment, the growth was performed both on plain SOI wafers and on patterned wafers. Patterning was performed by reactive ion etching. The device layer was etched to form square mesas of 5 to 100 μm in width, seperated by trenches ~4 μm wide. The trench depth was equal to the device layer thickness, therefore the trench exposed the underlying SiO_2 layer. In this way two goals were achieved: (*i*) complete mechanical separation between mesas; (*ii*) oxide faces in the trenches enable a selective growth only on Si mesas. Curvature analysis was performed by stylus profile measurements on the samples before and after the growth of GaN. Prior to growth, the SOI mesas revealed perfect planarity within the accuracy of measurement.

In Fig. 1, a GaN layer grown on an unpatterned SOI substrate is shown. Cracks cross the layer in a way similar to that observed on plain Si substrates, but here the characteristic distance between parallel cracks is up to one order of magnitude larger. Cracks in GaN grown on plain Si tend to penetrate into the substrate to a ~ 1 μm depth. When GaN is grown on SOI the crack penetration effect is further enhanced. It is clearly seen in Fig. 1 that the layer peels off together with the Si device layer, and one can observe the track of the cracking on the SiO_2 surface revealing the direction of crack propagation underneath the Si. The reduced crack density in this composite clearly demonstrates partial release of thermal strain.

Fig. 2(a) shows GaN islands grown on ~100 μm Si mesa. The islands are seen to bend upwards towards their perimeters. A close look at an island [Fig. 2(b)] shows that the GaN layer is fully attached to the Si mesas; however the BOX layer, underneath the mesa, was torn laterally, such that part of it is attached to the handle and the other part is attached to the Si mesa. The oxide attached to the Si mesa is of varying thickness. It is about 1 μm thick around the perimeter and approaches the full BOX thickness close to the mesa center. The three layers (GaN, device layer and BOX) are bent upwards together.

Fig. 3 shows a typical profile measurement taken on top of a GaN island. An average height difference of ~1.6 ± 0.1 μm between perimeter and island center was found, which corresponds to a radius of curvature of ~ 0.78 ± 0.07 mm.

In order to calculate the stress in the layers the model of Olsen and Ettenberg is used.[11] The model assumes force equilibrium

$$\sum_{i=1}^{3} F_i = 0 \tag{1}$$

and moment equilibrium for a measured curvature radius R,

$$\sum_{i=1}^{3}\left[\frac{E_i t_i^{\,3}}{12R} + F_i\left(\sum_{j=1}^{i} t_j - \frac{1}{2}t_i\right)\right] = 0. \tag{2}$$

where E_i is Young's modulus and t_i is the thickness of the ith layer. The strain at the interfaces between layers i and i+1 is calculated by

$$\varepsilon_i - \varepsilon_{i+1} = \frac{F_{i+1}}{E_{i+1} w t_{i+1}} - \frac{F_i}{E_i w t_i} - \frac{1}{2R}\left(t_i + t_{i+1}\right) \tag{3}$$

where w is the structure width. The stress in the ith layer, as a function of the distance z from its bottom surface, is

$$\sigma_i(z) = \left[\frac{F_i}{w t_i} + \frac{E_i}{R}\left(z - \frac{t_i}{2}\right)\right] \Big/ \left(1 - \nu_i\right) \tag{4}$$

where ν_i is the layer Poisson's ratio. In our structure i = 1, 2, 3 applies to the SiO_2, Si and GaN layers, respectively. The data used for calculation is shown in Table I. E and ν for $\{111\}_{Si}$ and for $\{0001\}_{GaN}$ planes were calculated from Si and GaN elastic constants, C_{ij} (Refs. 12 and 13, respectively). The relations used for calculation of GaN properties follow Refs. 14 and 15.

The above model was applied to our structure, knowing the radius of curvature at room temperature. The calculated stresses in the layers at the structure center are plotted in Fig. 4. The stress in the GaN layer is ~340 MPa at the GaN/Si interface, but it rapidly decreases to ~ –260 MPa at the surface. Thus, the GaN surface, being under compressive stress, apparently acts to prevent crack formation. For comparison, a 1.9 μm GaN grown on plain Si (~280 μm thick), after 1000°C cooling, suffers (if not cracked) ~480 MPa all across its thickness, but gets cracked long before reaching this cooling degree. Therefore, the growth on SOI mesas causes both the reduction of stress in the GaN/Si interface, and the compressive stress at the top surface. The first may explain the increase of distance between cracks in the unpatterned GaN/SOI while the latter may explain the lack of cracks in large GaN patterns on SOI.

The crack density reduction due to interfacial stress decrease, may be explained in terms of reduction of strain energy stored in the layered structure, such that less crack surfaces are needed to release the stored energy. Therefore the distance between cracks increases when GaN is grown on SOI.

The growth on patterned SOI enables large area crack free growth due to an additional factor. In order to completely avoid the formation of cracks, it is desirable to enable energy release without the need for any surface addition by cracking. This may be achieved by creating artificial surfaces, such as in LCE. A reduction of the lateral size by LCE creates free faces which act similarly to cracks in the layer. The free faces reduce the total area which suffers the highest stress in the GaN/Si interface, since the stress goes to zero towards free faces.[4, 16]

Using Eqs. (1), (2) and (3), and the measured value of R, the experimental values of interfacial strain $\varepsilon_i - \varepsilon_{i+1}$ were obtained for both interfaces, GaN/Si and Si/SiO_2. These values can be compared to the expected thermal strain

$$\varepsilon_i - \varepsilon_{i+1} = (T_f - T_g)(\alpha_i - \alpha_{i+1}) \tag{5}$$

where T_f is room temperature and T_g is the growth temperature and $\alpha_{i,i+1}$ are the layers' coefficients of thermal expansion. For a temperature difference of (T_g - T_f)~1000°C it was found that in both interfaces the measured strain is larger than the calculated thermal strain by ~32% and ~31%, respectively. This suggests that additional stresses exist in the structure, apart from the thermal stress. A possible large stress source in the GaN/Si interface may be the large lattice mismatch, which

even after relaxation by threading defects, maintains a residual stress. The large strain deviation in the Si/SiO_2 interface may be due to the SOI manufacturing process, which includes hot attachment of the device Si wafer to the top side of the handle wafer, which is covered with thermal oxide. It may also be introduced by the thinning process, which the device wafer underwent. Finally, it may be explained by thickness variations of the SiO_2 layer, since, as seen in Fig. 2(b), the SiO_2 thickness varies from edge to center.

The BOX lateral cracking, seen at Figs. 1 and 2(b), which originates at the edges of the mesa, occurs close to the center of the BOX thickness. This may be explained by the strong adhesion between the Si and the SiO_2, versus the lower SiO_2 bulk strength. The low BOX strength is also the main cause for the peeling observed in the unpatterned GaN on SOI (Fig. 1), however in the unpatterned layers the lateral cracks in the BOX may be initiated by the vertical cracks which cross the Si device layer.

In summary, GaN layers were grown on patterned and unpatterned SOI substrates. It was found that the growth on unpatterned SOI reduces the crack density, as compared to the growth on plain Si(111). Moreover, the use of patterned SOI enables the growth of large area, crack free GaN islands, which size is of an order of magnitude suitable for device applications. The GaN islands are strongly bent upwards. The SOI wafer compliance is, therefore, enhanced by Lateral Confined Epitaxy. Further work is to be done in order to control the bending for specific applications.

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REFERENCES

[1] D. M. Follstaedt, J. Han, P. Provencio and J. G. Fleming, *MRS Internet J. Nitride Semicond. Res.*, AS1, G3.72 (1999).

[2] A. Dadgar, A. Alam, T. Riemann, J. Bläsing, A. Diez, M. Poschnreider, M. Straßburg, M. Heuken, J. Christen and A. Krost, (to be published in *Phys. Stat. Sol. A*).

[3] H. Marchand, L. Zhao, N. Zhang, B. Moran, R. Coffie, U. K. Mishra, J. S. Speck, S. P. DenBaars, and J.A. Freitas, *J. Appl. Phys.*, **89** (12), 7846, (2001).

[4] S. Zamir, B. Meyler, and J. Salzman, *Appl. Phys. Lett.*, **78** (3), 288, (2000).

[5] S. Zamir, B. Meyler, and J. Salzman, *J. Crys. Growth*, **230** (3-4), 341, (2001).

[6] S. Zamir, B. Meyler, and J. Salzman, *(unpublished).*

[7] Z.Yang, F. Guarin, I. W. Tao, W. I. Wang, and S. S. Iyer, *J. Vac. Sci. Technol. B*, **13** (2), 789, (1995).

[8] A. J .Steckl, J. Devrajan, C. Tran and R. A. Stall, *Journal of Electronic Materials*, **6** (3), 217, (1997).

[9] J. Cao, D. Pavlidis, A. Eisenbach, A. Philippe, C. Bru-Chevallier and G. Guillot, *Appl. Phys. Lett.*, **71** (26), 3880, (1997).

[10] Usually a substrate is referred to as compliant if its thickness is of the order of the critical thickness to avoid mismatch dislocations in the epilayer to be grown. This is not the case here. The 2.3 μm thick Si(111) was chosen in order to study the partial strain release upon cooling, and its effect on thermal cracks.

[11] G. H. Olsen and M. Ettenberg, *J. Appl. Phys.*, **48** (6), 2543, (1977).

[12] W. A. Brantley, *J. Appl. Phys.*, **44** (1), 534, (1973).

[13] V.A. Savastenko, and A. U. Sheleg, *Phys. Stat. Sol. A.*, **48**, K135, (1978).

[14] J. F. Nye, *Physical Properties of Crystals*, (Clarendon Press, Oxford, 1957), Chap. VIII.

[15] R. Thokala, J. Chaudhuri, *Thin Solid Films*, **266**, 189, (1995). Note that $S_{11}=[2(C_{11}-C_{12})]^{-1}-SC_{33}/2$, where $S=[(C_{11}-C_{12})C_{13}-2C_{13}^2)]^{-1}$.

[16] J. H. Selverian and D. O'Neil, *Thin Solid Films*, **235**, 120, (1993).

LIST OF TABLES

TABLE I. Material parameters used for calculations.

Material	Coeff. of thermal expanssion α [10^{-6} K^{-1}]	Poisson's Ratio ν	Young's Modulus E [GPa]
GaN	5.59 [a, b]	0.18 [c]	195.9 [c]
Si	3.59 [b]	0.262 [d, e]	169.2 [d, e]
SiO_2	0.55 [f]	0.16 [f]	71.7 [f]

[a] S. Strite and H. Morkoç, *J. Vac. Sci. Technol. B*, **10** (4), (Jul/Aug 1992).

[b] A. Trampert, O. Brandt and K. Ploog in *Semiconductors and Semimetals*, Vol. 50, Edited by J. I. Pankove, and T. D. Moustakas, (Academic Press, San Diego, California, 1998), pp. 167-192.

[c] Calculated for $\{0001\}_{GaN}$ planes by data from Ref. 10.

[d] *CRC Handbook of Chemistry and Physics*, 75th Ed., (CRC Press, 1994), Chap. 12. p. 33.

[e] W. A. Brantley, *J. Appl. Phys.*, **44** (1), 534, (1973).

[f] *CRC Handbook of Chemistry and Physics*, 75th Ed., (CRC Press, 1994), Chap. 12, p. 188. (Taken for clear fused quartz).

FIGURE CAPTIONS

FIG. 1. An SEM micrograph of GaN layer on an unpatterned SOI substrate. Cracks cross the layer in a way similar to that observed on plain Si substrates. The layer peels off together with the top Si layer (arrowed region), and the track of the lateral cracking of the SiO_2 layer is revealed.

FIG. 2. (a) An SEM micrograph of GaN islands grown on ~100 μm SOI mesa. The islands are seen to bend upwards towards their perimeters. (b) A close look at an island edge showing the upward bending of the whole three layer composite.

FIG. 3. A typical profile measurement taken on top of a GaN island. An average height difference of ~1.6 ± 0.1 μm between perimeter and center was found, which corresponds to a radius of curvature of ~ 0.78 ± 0.07 mm.

FIG. 4. The stress distribution in the composite structure, calculated by the model.

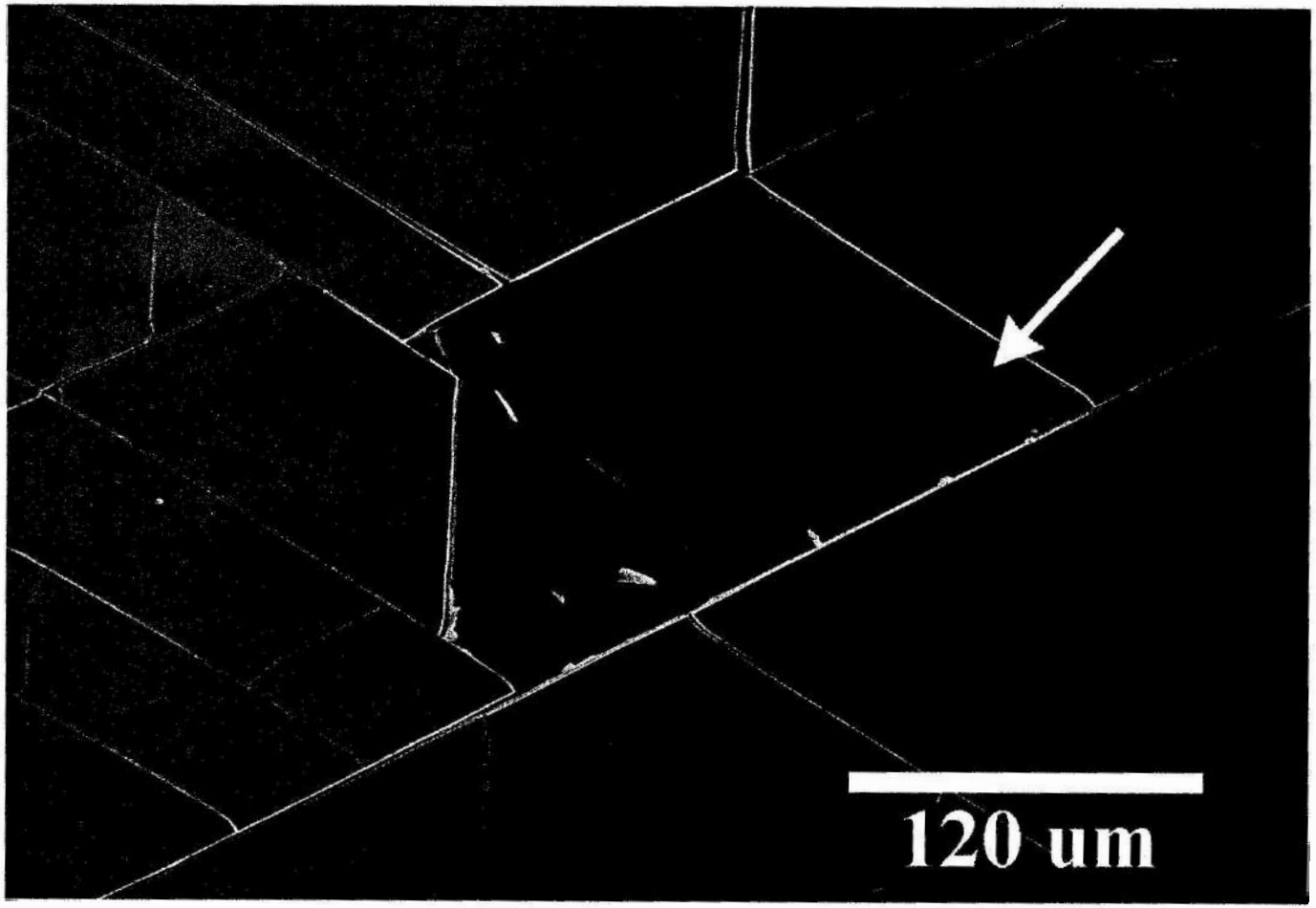


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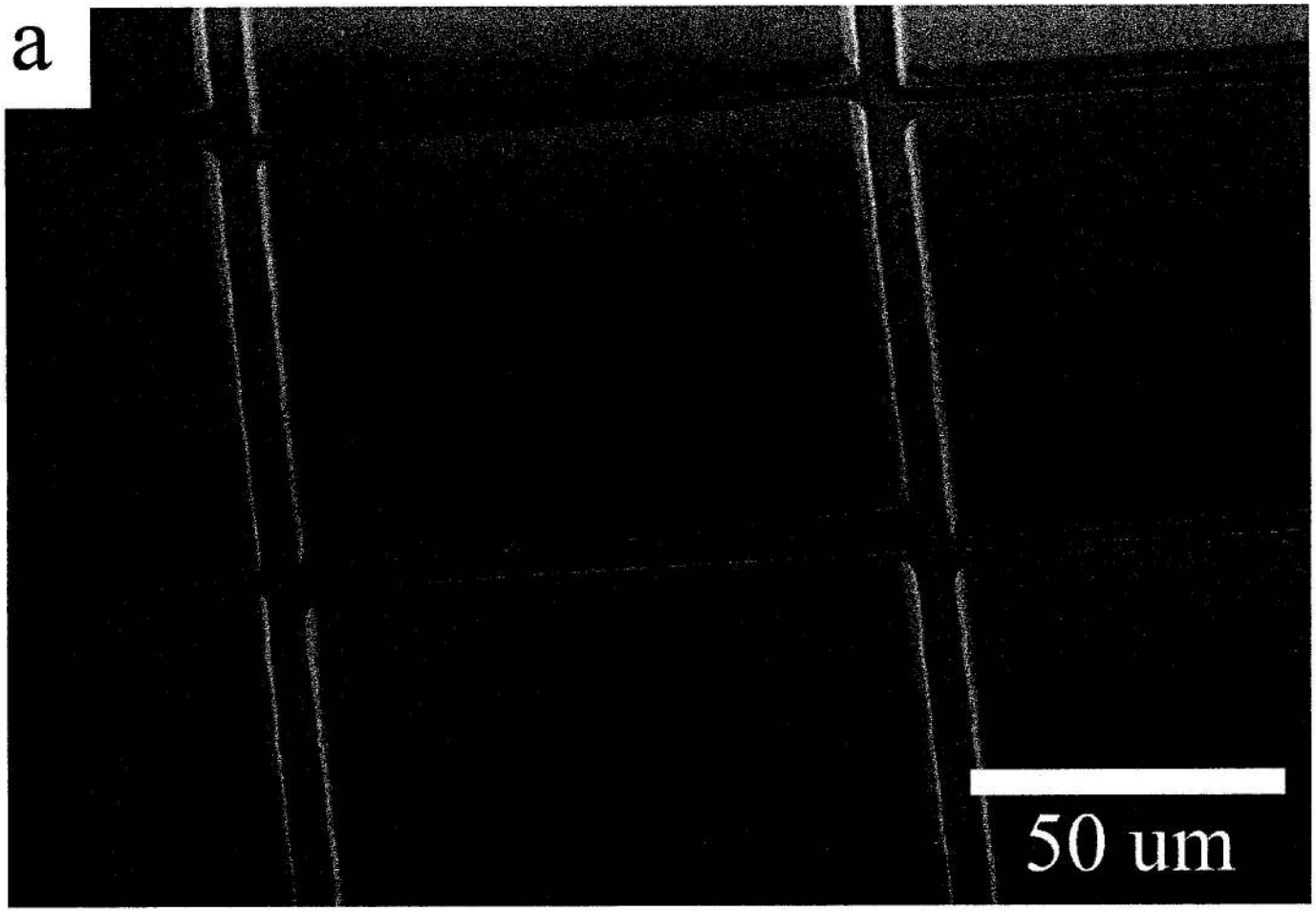


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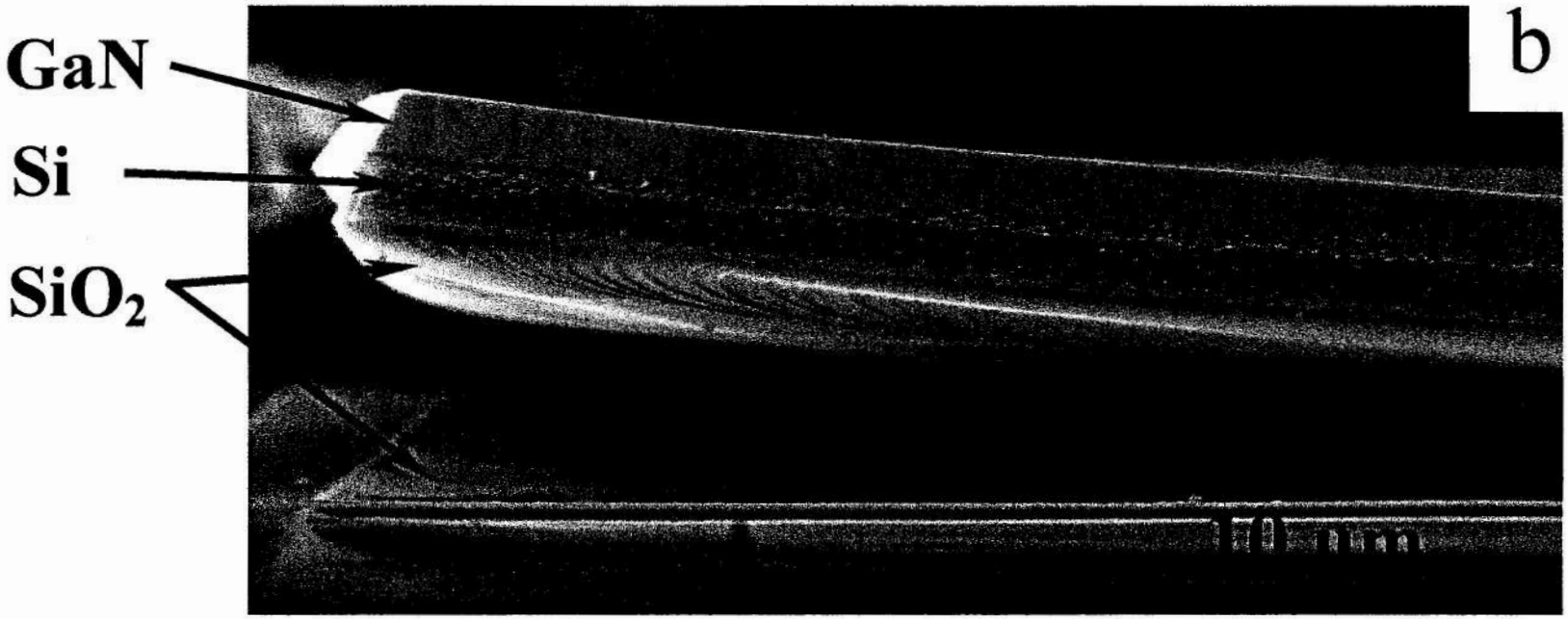


FIG. 2(b) of 4

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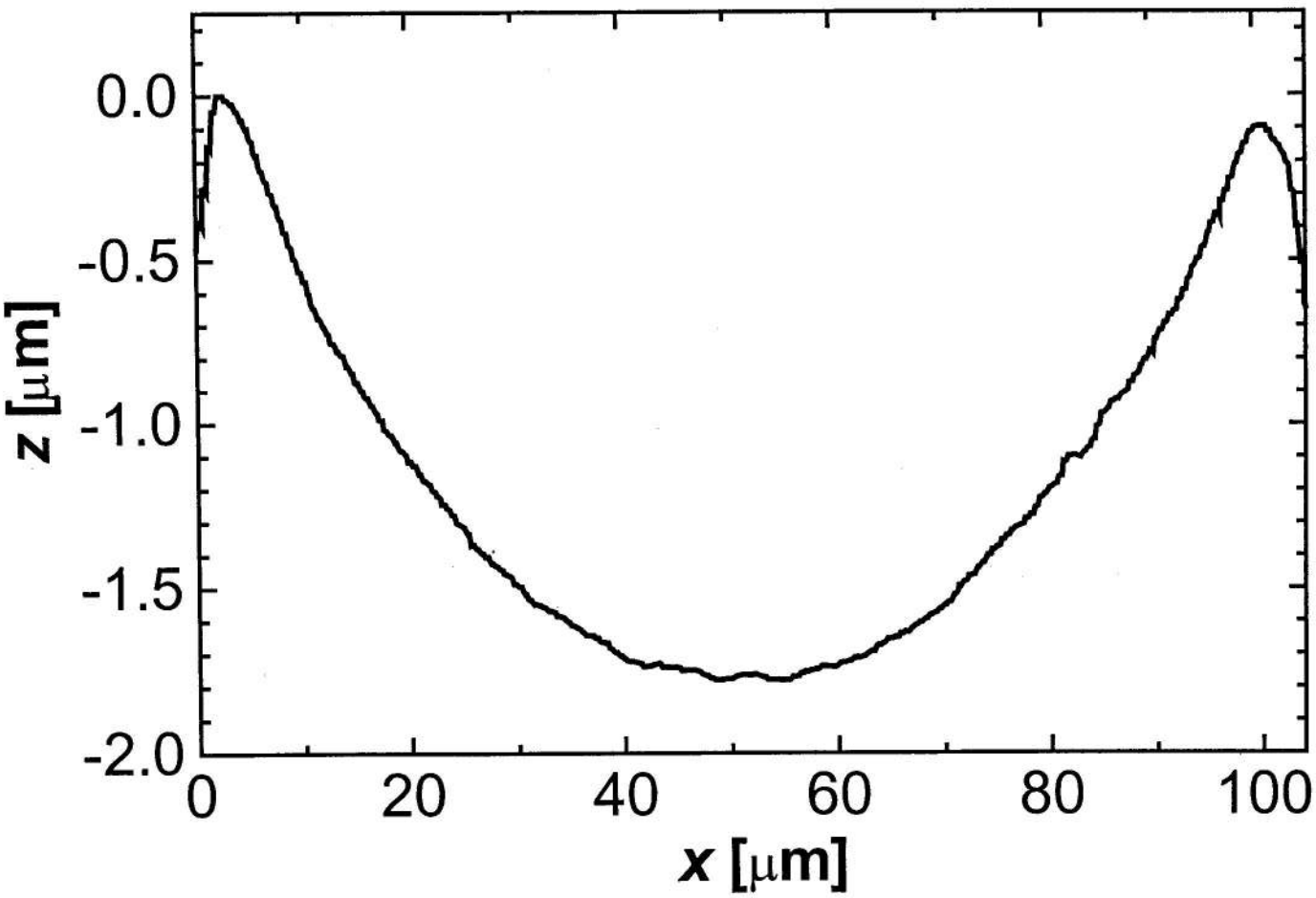


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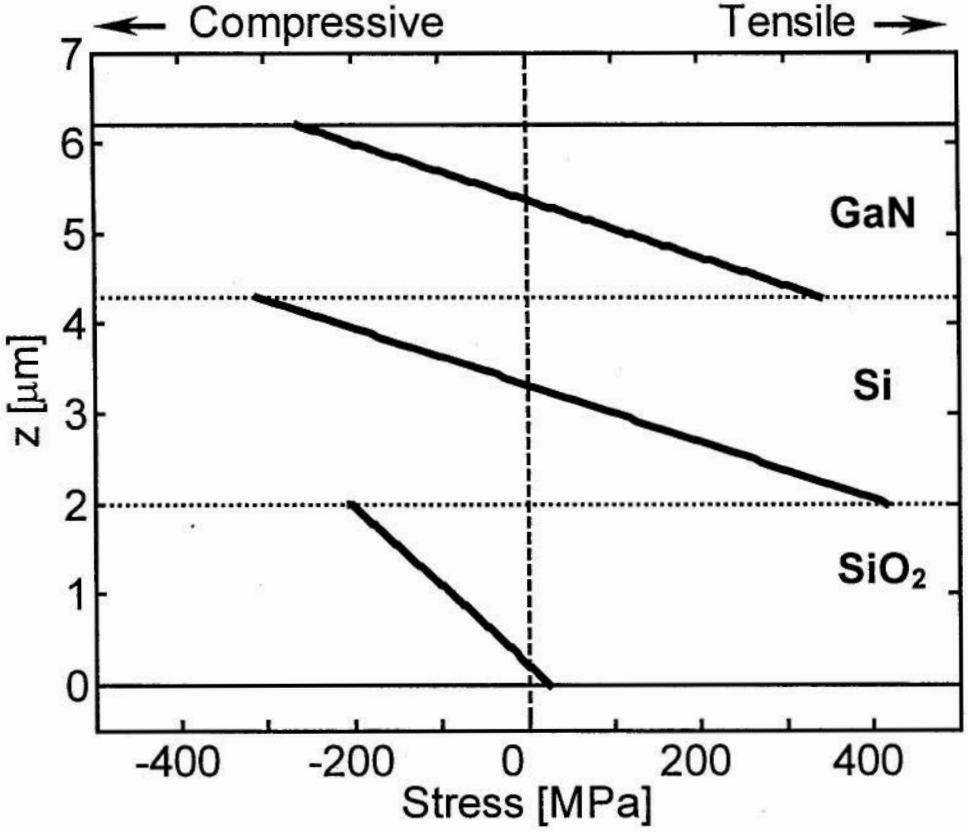


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